TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 9 OF 10

Draw and Measure



Big idea: Be exact to the nearest centimetre. A ruler counts centimetres. Start at the 0 mark, read the mark at the other end, and draw or measure a line exactly.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Length (Length in centimetres (cm); Measuring a line to the nearest cm; Drawing a line of a given length; Comparing lengths: how much longer) Strand label for the tutor only (MOE Measurement, length in cm, 1.1 to 1.4). It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as 'MOE-aligned'. Deeper than the holiday cubes-and-ruler lesson, which is warm-up only.
Objectives	• Use a ruler correctly: put the 0 mark at the start, keep it straight, read the mark at the other end. • Draw a straight line of a given whole number of centimetres, up to 14 cm. • Measure to the nearest centimetre: when the end is between two marks, choose the closer mark. • Compare two lengths and say how much longer one is. • Find the length of a line that starts at a mark other than 0, and of a bent path made of straight pieces.
Joining cold	**Remember warm-up covers:** Measure with same-size things laid end to end, no gaps and no overlaps, then count them; Find 0 on a ruler and count along the marks to 12; A centimetre (cm) is the gap from one mark to the next. Only counting to 20. A child who has never used a ruler is shown 0, the marks and the gaps with their own ruler in the warm-up. Nothing from earlier Term 1 lessons is needed. If the child has done the holiday measuring lesson, laying things end to end and reading cm will feel familiar. This lesson stands alone: it needs no earlier lesson except through the warm-up. It is one of the safer Term 1 lessons to miss (plan: the safest is Lesson 4). A child who misses it watches the recording and does workbook pages 1 and 2 with a ruler before Lesson 10.
Child materials	• A ruler with cm marks, starting at 0 (15 or 30 cm) • 3 or 4 things to measure; 10 paper clips (or small same-size things) • Workbook pages 1 to 5 (A4; page 6 is not used), a pencil, a rubber, spare paper
Tutor materials	• A real ruler, a pencil and a strip of paper to show on camera; a phone or camera that can look down at the table • A visible clock or a timer slide Families keep a ruler at home, so no ruler is printed in the workbook. If a child has none, a grown-up can look for any straight-edged ruler or strip marked in centimetres, or tell the tutor before the lesson. Plastic or wooden rulers only, no metal edges. Nothing needs to be bought. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears.
Tiers	Start here (7 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder only: a bent path: add the straight pieces; a line that starts at a mark other than 0 (end mark take away start mark); cutting a ribbon into pieces (the total stays the same). Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself with a real ruler first. Check that question 3 and the pencil examples work with your own pencil.
- Tell parents a week before: a ruler marked in centimetres starting at 0, workbook pages 1 to 5 printed on A4 (page 6, the cut-out page, is not used in this lesson), about 4 things to measure, 10 paper clips or small same-size things, a pencil and spare paper.
- Have your own ruler, a pencil and a strip of paper ready, and a camera that can look down at the table. Have the slides open and check the clock or timer is visible on screen share.
- Check the recording settings exclude children's video. If a family has no ruler, ask them before the lesson so a grown-up can find one; do not ask the child in front of the group.

Safeguarding and recording (this lesson)

- Rulers stay flat on the table when not in use: no swinging, no flicking, no pointing. Plastic or wooden rulers only, no metal edges, no compasses. Small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Sneak-peek and exit answers are shown on the child's own paper to the camera at the same moment, once. There is no private chat and no board held up in front of the group. When asked to show a ruler or a thing, the child holds it to the camera: the tutor looks at the ruler and the paper, not at the child.
- Never read out a child's name with a right or wrong answer. Never show a child's page in the recording; record the tutor and the screen only, with children's video excluded.
- A child who ticks "I am still stuck" gets a check-in in the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby). No private messages to children.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–4	**Opening**	4	1: Draw and Measure	2:58
4–10	**Remember warm-up**	6	3: Warm-up: measure with clips, Find 0 on your ruler, One centimetre	5:06
10–14	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:42
14–28	**Explore**	14	3: Explore: guess, then measure, Draw it!, What did we notice?	11:42
28–38	**Mini-lesson**	10	7: The parts of a ruler, Measure: three steps, Draw a line 7 cm long, Between two marks, A line that starts at 3, Your turn, Check: your turn	8:31
38–42	**Brain break**	4	1: Jump the tens, hop the ones	3:32
42–70	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	26:42
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	5:27
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:28
82–85	**Look-back**	3	1: Look-back	2:28
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:38
	Total	**90**		**75:13**

The script needs 75:13 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 89:08. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 14 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: the quick questions in Remember; the 12 cm line in Explore round two; the quick questions in the mini-lesson; the optional slide. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 4:00 Opening

script 2:58 of 4:00

0:00	SLIDE	Show slide **Draw and Measure** (s-title)
0:25	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your ruler. Pause or rewind any time.*
0:36	SAY	Good morning, inquirers! Cameras on, pencil ready, and your ruler on the table. This lesson is recorded so that a friend who is away can catch up. We record me and the screen only. You may switch your camera off whenever you like.
0:57	DO	Greet each child by first name. Ask everyone to hold their ruler up to the camera for a moment so you can see the 0 end. Remind them: a ruler stays flat on the table, and it is only for measuring. [0:55]
1:52	SAY	Here is a question to start. How long is a pencil? Show me with your two hands, as far apart as you think. Hold still.
2:04	CHILDREN	Children show a guess with their hands. Say what you see in general (some wide, some narrow). Nobody is right or wrong yet. [0:35]
2:40	SAY	Hands are not very exact. My hands and your hands are not the same size. Today we learn to measure and to draw lines exactly, so that everybody gets the same answer.
2:56	SAY	Let us warm up.
	NOTE	Ruler check: if a child has no ruler, a grown-up can look for any straight-edged ruler or strip marked in centimetres; do not ask the child to guess. No names on screen. Keep the greeting short and warm. Tell the group again that recordings show only you and the screen.

4:00 – 10:00 Remember warm-up

script 5:06 of 6:00

4:00	SLIDE	Show slide **Warm-up: measure with clips** (s-rem1)
4:25	REC	*Recording: join in! Lay clips along a pencil, find 0 on your ruler, and count the marks out loud. Pause if you need longer.*
4:37	SAY	Get your pencil and your paper clips, or small things that are all the same size. Lay them end to end along your pencil. No gaps. No overlaps. Then count them.
4:52	CHILDREN	Children lay clips (or pasta, short straws) along a pencil and count them. [0:50]
5:42	ASK	How many clips long is your pencil? Say your number or show it on your fingers. [wait about 20 s for answers]
6:10	SAY	My pencil is 6 big clips long. Now I use small clips: 9. Same pencil, different numbers! The clips are not the same size. So people agree to use one unit that is the same for everybody: the centimetre.
6:30	SLIDE	Show slide **Find 0 on your ruler** (s-rem2)
6:55	SAY	Now your ruler. Find the 0 mark. Put your finger on it. Count along the marks with me: 0, 1, 2, 3, all the way to 10.
7:08	CHILDREN	Count the marks together, pointing with a finger, from 0 to 10, then on to 12. [0:30]
7:38	ASK	Which mark comes just after 4? [wait about 10 s for answers]
7:52	ASK	Which mark comes just before 10? [wait about 10 s for answers]
8:04	SLIDE	Show slide **One centimetre** (s-rem3)

8:30	SAY	The space between two marks is one centimetre. We write it cm. From 0 to 1 is 1 cm. Slide your finger from 0 to 5. Five gaps: that is 5 cm.
8:46	ASK	Slide your finger from 0 to 8. How many centimetres is that? [wait about 15 s for answers]
	IF STUCK	A child who counts the marks instead of the gaps (says 6 for 0 to 5): put a finger on the 0 and say zero, then count each jump aloud: one, two, three, four, five. The 0 is where we start, it is not a centimetre.
	NOTE	Answers: any clip count is fine (the number depends on the clips). Then 5, 9, 8. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20 and a ruler.

10:00 – 14:00 Sneak-peek puzzles

script 3:42 of 4:00

10:00	SLIDE	Show slide **Sneak-peek puzzles** (s-sneak)
10:25	REC	*Recording: pause now and try both puzzles on your own paper. Guessing is fine. Press play for the next part.*
10:35	SAY	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
10:48	READ	Puzzle one. Draw a dot. Use your ruler to draw a line from the dot, exactly 3 cm long.
10:58	CHILDREN	Children draw the dot and the 3 cm line on their own paper. [0:55]
11:53	READ	Puzzle two. Ali puts one end of a pencil at the 1 mark of his ruler. The other end is at the 9 mark. Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.
12:15	CHILDREN	Children write or draw their answer. [0:55]
13:10	HOLD UP	Everyone holds their paper to the camera at the same moment, once, after puzzle two. Do not read answers aloud. No private chat.
13:35	SAY	Thank you. We will come back to puzzle two in a few minutes.
	NOTE	Puzzle 1 answer: a line 3 cm long (the 0 mark at the dot, the 3 mark at the other end). Puzzle 2 answer: no, 8 cm, because the pencil starts at the 1 mark. Look for good ideas to praise later (a child who started at 0, or who noticed the 1). Do not write scores or keep a list.

14:00 – 28:00 Explore

script 11:42 of 14:00

14:00	SLIDE	Show slide **Explore: guess, then measure** (s-explore1)
14:25	REC	*Recording: pause now. Guess, then measure three things. Then draw lines of 3, 8 and 12 centimetres. Press play when you have tried.*
14:36	SAY	Today your ruler is your teacher. Find three things: a pencil, a spoon, a rubber or a toy. First guess how many centimetres long each one is. Then measure.
14:51	SAY	Put one end at the 0 mark, not at the edge of the ruler. Read the mark at the other end. If the end is between two marks, look which mark is closer.
15:08	CHILDREN	Round 1: each child guesses, then measures three things. Tutor scans the cameras and asks a few children to hold their thing and ruler to the camera: look at the 0 end. [3:50]
18:58	ASK	Tell me about one of your things. How long was it? How close was your guess? [wait about 50 s for answers]
19:56	SLIDE	Show slide **Draw it!** (s-explore2)
20:20	SAY	Round two: draw it! Put a dot. Put the 0 mark on the dot. Make a small mark at the end. Join the dot and the mark with a straight line. First, draw a line 3 cm long.
20:40	CHILDREN	Children draw 3 cm. [0:35]
21:14	SAY	Now draw a line 8 cm long.
21:18	CHILDREN	Children draw 8 cm. [0:40]

21:58	**SAY**	Now a line 12 cm long.
22:01	**CHILDREN**	Children draw 12 cm. [0:45]
22:46	**HOLD UP**	On a count of three, everyone holds their paper to the camera. Look at where the lines start. No answers read aloud.
23:11	**ASK**	Ravi started his line at the very edge of his ruler, not at the 0 mark, and stopped at the 8 mark. Is his line exactly 8 cm? How do you know? [wait about 40 s for answers]
24:07	**SLIDE**	Show slide **What did we notice?** (s-explore3)
24:32	**ASK**	What did you notice? Where do we start measuring? What do we do if the end is between two marks? [wait about 45 s for answers]
25:27	**SAY**	We start at the 0 mark. We read the mark at the other end. If the end is between two marks, we choose the closer one. That is the nearest centimetre.
	IF STUCK	A child who measures from the edge of the ruler: ask them to find the 0 mark and slide the thing along until its end is on the 0. A child with a 15 cm ruler measuring something longer: ask them to measure a shorter thing, or mark the end of the ruler with a finger and slide on.
	IF STUCK	A child with nothing to measure: any straight line works, such as the edge of a book or a line they draw on paper.
	NOTE	Main room only: no breakout rooms in this lesson, so you can see every camera. Pairs are optional: a grown-up may check the 0 end. Ravi answer: no. He began a little before the 0 mark, so his line is a little longer than 8 cm. Praise the reason, not the speed.

28:00 – 38:00 Mini-lesson

script 8:31 of 10:00

28:00	**SLIDE**	Show slide **The parts of a ruler** (s-m1)
28:25	**REC**	*Recording: keep watching and copy each example with your own ruler. Rewind if you miss a step.*
28:34	**SAY**	Look at your ruler. The numbers sit on the marks. 0 is where we start. The gap from one mark to the next is one centimetre.
28:46	**ASK**	How many centimetres from the 0 mark to the 6 mark? Slide your finger and say it. [wait about 12 s for answers]
29:07	**SLIDE**	Show slide **Measure: three steps** (s-m2)
29:32	**SAY**	To measure: one end at 0, the ruler straight, and read the mark at the other end. Watch my pencil.
29:42	**DO**	Show on camera: a pencil with its end exactly on the 0 mark, the ruler straight, read the tip aloud. Say the three steps while you do them. [0:35]
30:17	**SLIDE**	Show slide **Draw a line 7 cm long** (s-m3)
30:42	**SAY**	To draw a line 7 cm long: a dot for the start. The 0 mark on the dot. A small mark at the 7. Then join them with a straight line. I draw while you watch.
31:00	**DO**	Draw a 7 cm line on paper under the camera, saying each step. Check at the end: 0 at the start, 7 at the end. [0:40]
31:40	**SLIDE**	Show slide **Between two marks** (s-m4)
32:05	**SAY**	Sometimes the end is not on a mark. Here it is between 8 and 9, and closer to 9. We say about 9 centimetres: the nearest centimetre.
32:18	**ASK**	Another one: between 12 and 13, and closer to 12. How long, to the nearest centimetre? [wait about 15 s for answers]
32:42	**SLIDE**	Show slide **A line that starts at 3** (s-m5)
33:06	**SAY**	A line that does not start at 0. It starts at the 3 mark and ends at the 8 mark. Count the jumps: 1, 2, 3, 4, 5. It is 5 cm long, not 8!
33:24	**ASK**	Why is it 5 cm and not 8 cm? [wait about 20 s for answers]

33:48	SLIDE	Show slide **Your turn** (s-m6)
34:14	READ	Your turn. Draw a line 5 cm long. Then: a line goes from the 2 mark to the 9 mark. How long is it? Write both answers on your paper.
34:30	CHILDREN	Children draw 5 cm and work out the second line. [1:20]
35:50	SLIDE	Show slide **Check: your turn** (s-m6a)
36:15	READ	The 5 cm line starts at 0 and ends at 5. The other line is 7 cm: count 7 jumps from 2 to 9, or take 2 from 9.
	NOTE	Key words every time: ruler, mark, centimetre (cm), start at 0, nearest. Do not teach millimetres or metres. Do not call the 0 a centimetre. The jump idea (count the gaps) is the same as hopping along a number line.
	IF STUCK	If a child draws from the end of the ruler: ask them to put a finger on the 0 mark and slide the ruler until the 0 is on the dot. If a child says the 2-to-9 line is 9 cm: ask them to count the jumps from the 2.

38:00 – 42:00 Brain break

script 3:32 of 4:00

38:00	SLIDE	Show slide **Jump the tens, hop the ones** (s-break)
38:25	REC	*Recording: stand up and join in! Big jump for ten centimetres, small hop for one.*
38:32	SAY	Stand up and get a sip of water. We move our bodies today. Let us pretend. A big jump is 10 cm. A small hop is 1 cm. So 14 cm is one big jump and four small hops.
38:52	CHILDREN	Call 14, 23, 30, 8. Children jump the tens and hop the ones. Count the jumps and hops aloud together. [2:40]

42:00 – 70:00 Workbook

script 26:42 of 28:00

42:00	SLIDE	Show slide **Workbook page 1: Start here** (s-ws1)
42:25	REC	*Recording: pause now and do workbook page 1 with your ruler. When you finish a page, press play to check it.*
42:36	READ	Page 1, Start here. Question 1: draw a line 7 centimetres long. Start it at the 0 mark. Question 2: draw a line 4 cm long and a line 9 cm long. Which is longer, and by how much? Question 3: measure your pencil. Write your guess first.
43:02	CHILDREN	Page 1, questions 1 to 3. Tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. [9:00]
52:02	SLIDE	Show slide **Check page 1** (s-ws1ans)
52:27	READ	Question 1: your line ends at the 7 mark. Question 2: the longer line is 9 cm, and it is 5 cm longer. Question 3: any measure close to your guess is fine. Mark your own page in another colour.
52:48	SAY	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
52:58	SLIDE	Show slide **Workbook page 2: Start here** (s-ws2)
53:22	READ	Page 2. Question 4: Ali and Mei measure pencils. Question 5: draw a line 5 cm long, then a line 3 cm longer than the first. Question 6: Mei's line goes from the 0 mark to the 7 mark. Mei says it is 8 cm long. Is she correct? Question 7: the tip of Ben's spoon is between two marks.
53:55	CHILDREN	Page 2, questions 4 to 7. [9:00]
62:55	SLIDE	Show slide **Check page 2** (s-ws2ans)
63:20	READ	Question 4: 13 cm, 9 cm, and 4 cm longer. Question 5: 8 cm. Question 6: no, her line is 7 cm. Question 7: 15 cm.
63:34	SLIDE	Show slide **Choose your next page** (s-choice)
63:59	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time.

64:12	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [4:30]
	IF STUCK	Q1 and Q2 (drawing): put a finger on the 0 mark, slide the ruler to the dot. Q3: guess first, then measure; a guess that is far off is fine. Q6: ask where her line ends on the ruler. Q7: which mark is the tip closer to?
	IF STUCK	A child who is still stuck ticks the stuck box on the look-back page. You check in during the open session in the next lesson (a quick visit to the shared room or a hand signal, parent nearby). Never a private message.
	NOTE	Think harder is for ideas taught today in a new setting. Go further is for mastery and extension. Do not announce who is on which page. Pairs are never split unless both children agree.

70:00 – 76:00 Talk time

script 5:27 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*
70:32	READ	Mei measures a pencil and says it is 15 cm. Ben measures the same pencil and says it is 16 cm. Why might their answers be different? How can we check?
70:49	CHILDREN	Think for a moment. Put a pencil next to your ruler and show your screen what you would do. [0:50]
71:39	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [1:00]
72:39	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers]
74:15	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
74:40	SAY	Remember puzzle two? Ali put his pencil at the 1 mark and the other end was at the 9 mark. He said 9 cm. The pencil is 8 cm long, because it starts at the 1, not at the 0.
75:00	ASK	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers]
	NOTE	Model answer: the tip may be between the 15 and the 16 mark, so one child chose the other mark; or one child did not start at the 0 mark. Check by measuring again with the 0 at the start and choosing the closer mark. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:28 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:25	REC	*Recording: pause and do the exit ticket on your own, with your ruler. Then press play to check it.*
76:34	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5. You may use your ruler.
76:45	CHILDREN	Children answer E1 to E3 on their own. [2:30]
79:15	HOLD UP	Everyone holds their ticket to the camera at the same moment so only you see it. No private chat.
79:40	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:05	READ	Question 1: your line starts at 0 and ends at 6. Question 2: 9 centimetres longer. Question 3: 7 centimetres, because 10 take away 3 is 7.
80:20	SAY	Check your own ticket in a different colour. Well done. I will stamp your inquiry journal.
	NOTE	Mark S, D or H in the tutor tracker (tutor only): S = secure on all three, D = developing (one wrong), H = needs help (two or three wrong). If D or H, plan two minutes of catch-up at the start of Lesson 10, in the open session, no names. Do not read out any child's result.

82:00 – 85:00 Look-back

script 2:28 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out.
82:48	**CHILDREN**	Children fill in the look-back box on page 5. [1:40]

85:00 – 90:00 Flex time and goodbye

script 0:38 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson is Story Steps: Read It Twice. You need only a pencil and paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: how many cuts?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: a strip of paper is 10 cm long. You may cut it at the 1 mark, the 2 mark and so on, but not at the ends. How many places can you cut? And if the strip is 20 cm long?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 9 places for the 10 cm strip, 19 for the 20 cm strip (one fewer than the length, because we cannot cut at the two ends). If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask "How do you know?" and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Mei says a line that ends at the 9 mark is always 9 cm long. Is that always true? Draw lines that end at the 9 mark and start at 0, at 2 and at 4. Measure each one.	*Recording: pause. Draw 3 lines that all end at the 9 mark: one starts at 0, one at 2, one at 4. How long is each?*	Not always. A line that starts at 0 and ends at 9 is 9 cm. Starting at 2 it is 7 cm, and starting at 4 it is 5 cm. It is only true when the line starts at the 0 mark.
Make a puzzle for your grown-up	4 (optional)	Draw a path with 3 straight pieces. Write the total length on the back, but not the pieces. Give it to your grown-up to measure each piece and add them. Do they get your total?	*Recording: pause. Draw a path with 3 straight pieces. Ask a grown-up to measure and add them. Does it match your total?*	No single answer. The measured pieces should add up to the total on the back, give or take 1 cm from reading the nearest centimetre.

Teach it back	3 (optional)	Pretend your grown-up or a toy has never used a ruler. Teach them how to draw a line 9 cm long, step by step, and say why the 0 mark goes on the dot.	*Recording: pause. Teach a grown-up how to draw a line 9 cm long with a ruler, in steps.*	A dot for the start. The 0 mark on the dot. A small mark at the 9. Join them with a straight line. Check: 0 at the start, 9 at the end. We start at 0 because the centimetres are counted from the 0 mark.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 10 to 14)

Puzzle 1 (access): Draw a dot. Use your ruler to draw a line from the dot, exactly 3 cm long.

> **Answer: A line from the dot that is 3 cm long: the 0 mark on the dot and the 3 mark at the other end.**

Puzzle 2 (think): Ali puts one end of a pencil at the 1 mark of his ruler. The other end is at the 9 mark. Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.

> **Answer: No. It is 8 cm long. The pencil starts at the 1 mark, not the 0 mark, so the first centimetre of the ruler is not part of the pencil (9 take away 1 is 8).**

Unmarked. Papers are shown to the camera once, together. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **In the box, draw a line 7 cm long. Start at the 0 mark.**

The line starts at the ___ mark and ends at the ___ mark.

> **Answer: A line from the 0 mark to the 7 mark: it starts at 0, ends at the 7 mark and is 7 cm long**

Hint (say this first): Make a dot. Put the 0 mark on the dot. Make a small mark at the 7. Join them with a straight line.
Worked solution: Put a dot for the start. Line up the 0 mark with the dot. Make a small mark at the 7. Join the dot and the mark with a straight line. The line ends at the 7 mark, so it is 7 cm long.
Common mistakes: Starting at the edge of the ruler instead of the 0 mark (the line comes out a little too long).; Drawing to the 8 mark because the 0 was counted as one.
Time for a middle child: about 2.5 min.

S2 **Draw a line 4 cm long. Under it, draw a line 9 cm long.**

a) The longer line is ___ cm long.
b) It is ___ cm longer than the shorter line.

> **Answer: 9 cm; it is 5 cm longer**

Hint (say this first): Draw the short line first, then the long one under it. To see how much longer, count the cm from 4 to 9.
Worked solution: The 9 cm line is the longer one. To find how much longer, count on from 4 to 9: that is 5 jumps. So it is 5 cm longer (9 take away 4 is 5).
Common mistakes: Writing 13 by adding the two lengths.; Starting the 9 cm line at a mark other than 0.
Time for a middle child: about 3 min.

S3 **Measure the length of your pencil. Write your guess first. Then measure to the nearest cm.**

a) My guess: ___ cm
b) I measured: ___ cm

> **Answer: OPEN ANSWER, accept any sensible measure (guess and measure within about 3 cm of each other); a new pencil is about 17 to 19 cm. Check: the 0 mark at one end, read the mark at the other end**

Hint (say this first): Put one end of the pencil at the 0 mark. Keep the ruler straight. Read the mark at the other end. If it is between two marks, choose the closer one.
Worked solution: Put one end of the pencil on the 0 mark and keep the ruler straight. Read the mark at the other end. A new pencil is usually between 17 and 19 cm. If the tip is between two marks, write the closer one.
Common mistakes: Starting at the edge of the ruler instead of the 0 mark.; Writing the guess in the measure space, or leaving off the unit.
Time for a middle child: about 3 min.

S4 **Ali's pencil and Mei's pencil both start at the 0 mark of a ruler. Ali's tip is at the 13 mark. Mei's tip is at the 9 mark.**

a) Ali's pencil is ___ cm long.
b) Mei's pencil is ___ cm long.
c) Ali's pencil is ___ cm longer than Mei's.

Answer: 13 cm; 9 cm; Ali's is 4 cm longer

Hint (say this first): Both pencils start at 0, so the mark at the tip is the length. To find how much longer, count on from 9 to 13.
Worked solution: Both start at the 0 mark, so each tip mark is the length: Ali 13 cm, Mei 9 cm. From 9 to 13 is 4 jumps (13 take away 9 is 4), so Ali's pencil is 4 cm longer.
Common mistakes: Adding 13 + 9 = 22.; Writing 9 as the difference (that is the length of Mei's pencil).
Time for a middle child: about 2 min.

S5 **Draw a line 5 cm long. Under it, draw a line 3 cm longer than the first line.**

The second line is ___ cm long.

Answer: The second line is 8 cm long (5 + 3)

Hint (say this first): The second line is the first line and 3 more. How long is that? Draw it with the 0 mark at the start.
Worked solution: The first line is 5 cm. 3 cm longer is 5 + 3 = 8 cm. Draw it from the 0 mark to the 8 mark, under the first line.
Common mistakes: Drawing a line 3 cm long.; Starting the second line at the edge of the ruler, so it is a little too long.
Time for a middle child: about 3 min.

S6 **Mei's line goes from the 0 mark to the 7 mark. Mei says it is 8 cm long.**

a) Is Mei correct? Circle Yes or No.
b) Her line is ___ cm long.

Answer: No. Her line is 7 cm. She should draw on to the 8 mark

Hint (say this first): Where does her line start and end on the ruler? Read the end mark.
Worked solution: Mei started at 0 and stopped at the 7 mark, so her line is 7 cm long. A line 8 cm long must end at the 8 mark. She should draw on to the 8 mark.
Common mistakes: Saying yes because Mei said 8.; Counting the 0 as one, so the 7 mark looked like 8.
Time for a middle child: about 2.5 min.

S7 **The tip of Ben's spoon is between the 14 mark and the 15 mark. It is closer to the 15 mark. The spoon starts at the 0 mark.**

To the nearest cm, the spoon is ___ cm long.

Answer: 15 cm (the tip is closer to the 15 mark)

Hint (say this first): Which mark is the tip closer to? That mark is the length to the nearest centimetre.
Worked solution: The tip is between 14 and 15 and it is closer to 15. So, to the nearest centimetre, the spoon is 15 cm long.
Common mistakes: Writing 14 because the tip has not reached 15.; Writing 14 and a half.
Time for a middle child: about 2 min.

Think harder (page 3): the taught idea in a new setting

T1 **Use your ruler to draw a path with 3 straight pieces: 5 cm, 3 cm and 4 cm.**

The whole path is ___ cm long.

Answer: 12 cm (5 + 3 + 4)

Hint (say this first): Draw one piece at a time. Turn a corner and draw the next piece. Then add the three lengths.
Worked solution: Draw 5 cm, turn, draw 3 cm, turn, draw 4 cm. The path is as long as all three pieces together: 5 + 3 = 8, and 8 + 4 = 12 cm.
Common mistakes: Measuring in a straight line from the start to the end of the path.; Drawing one long line instead of turning at the corners.
Time for a middle child: about 5 min.

T2 **Line A starts at the 2 mark and ends at the 9 mark. Line B starts at the 3 mark and is 6 cm long.**

a) Line A is ___ cm long.
b) Line B ends at the ___ mark.

Answer: 7 cm; the 9 mark

Hint (say this first): Count the jumps from the start mark to the end mark. For line B, jump 6 along from the 3 mark.
Worked solution: (a) From the 2 mark to the 9 mark is 7 jumps: 9 take away 2 is 7 cm, not 9 cm. (b) Start at the 3 mark and jump on 6: 3 + 6 = 9. The line ends at the 9 mark.
Common mistakes: Writing 9 for line A because the end is at the 9 mark.; Taking 6 away from 3, or writing 6 for line B.
Time for a middle child: about 4 min.

T3 **Draw a line 13 cm long in the box. Put a dot on it 5 cm from the start.**

The dot is ___ cm from the end of the line.

Answer: 8 cm (13 take away 5)

Hint (say this first): The whole line is 13 cm. The first part is 5 cm. How many cm are left? You can check by measuring from the dot to the end.
Worked solution: The whole line is 13 cm and the part before the dot is 5 cm. The rest is 13 take away 5, which is 8 cm. Check by measuring from the dot to the end of the line.
Common mistakes: Writing 5, the part that is already known.; Adding 13 + 5 = 18.
Time for a middle child: about 4 min.

T4 **Both pencils start at 0. Ali's tip is between 16 and 17, closer to 16. Mei's tip is between 18 and 19, closer to 19.**

a) To the nearest cm, Ali's pencil is ___ cm and Mei's pencil is ___ cm long.
b) Mei's pencil is ___ cm longer than Ali's.

Answer: 16 cm; 19 cm; 3 cm longer

Hint (say this first): For each pencil, choose the mark that the tip is closer to. Then find how much longer by counting on from 16 to 19.
Worked solution: Ali's tip is closer to 16, so 16 cm. Mei's tip is closer to 19, so 19 cm. From 16 to 19 is 3 jumps, so Mei's pencil is 3 cm longer.
Common mistakes: Choosing the mark the tip has already passed (17 and 18).; Writing 2 by using 17 and 19.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **A ribbon is 30 cm long. Mei cuts all of it into 3 pieces: A, B and C. Piece A is 8 cm long. Piece B is 3 cm longer than piece A. How long is piece C?**

Piece C is ___ cm long.

Answer: Piece C is 11 cm (B is 11 cm, A and B together are 19 cm, and 30 take away 19 is 11)

Hint (say this first): Find piece B first. Then add A and B. How much of the 30 cm is left for C?
Worked solution: B is 8 + 3 = 11 cm. A and B together are 8 + 11 = 19 cm. The ribbon is 30 cm, so C is 30 take away 19, which is 11 cm. Check: 8 + 11 + 11 = 30.
Common mistakes: Taking 30 - 8 = 22 and stopping.; Writing 3 for piece B instead of 3 more than A.
Time for a middle child: about 5 min.

G2 **A snail starts at the 4 mark of a long ruler. It crawls to the 13 mark. Then it crawls back 5 cm towards the 0 mark.**

a) The snail is now at the ___ mark.
b) In all, the snail has crawled ___ cm.

> **Answer: At the 8 mark; it crawled 14 cm in all (9 cm forward and 5 cm back)**

Hint (say this first): Forward: from 4 to 13 is how many cm? Back: take 5 from the 13 mark. For the total, add the way forward and the way back.
Worked solution: Forward from the 4 mark to the 13 mark is 9 cm. Back 5 cm: 13 take away 5 puts the snail at the 8 mark. In all it crawled 9 + 5 = 14 cm, although it is only 4 cm from where it began.
Common mistakes: Writing 9 for the total and forgetting the way back.; Writing 4 as the mark (the distance from the start, not the mark).
Time for a middle child: about 6 min.

G3 **Draw a path with 3 straight pieces. The pieces are all different lengths, in whole cm. The whole path is exactly 20 cm long. Many answers are right.**

My 3 lengths: ___ cm, ___ cm and ___ cm.

> **Answer: OPEN ANSWER, accept any 3 different whole numbers that add up to 20, for example 5, 7 and 8 (5 + 7 + 8 = 20)**

Hint (say this first): Pick 3 different numbers and add them. Too small? Make one longer. Too big? Make one shorter. Then draw each piece.
Worked solution: One answer is 5, 7 and 8: 5 + 7 is 12 and 12 + 8 is 20. Other answers are 3, 8 and 9, or 4, 7 and 9. If the total is not 20, change one piece until it is.
Common mistakes: Two pieces the same length (for example 5, 5 and 10).; A total of 19 or 21 because a piece was drawn a little too long or too short.
Time for a middle child: about 7 min.

Investigation (page 5): Does the start mark matter?

Easy first step: Draw a line 6 cm long. Start it at the 2 mark of your ruler. Which mark does it end at?
Then: Try 3 more start marks. Each time, draw a line 6 cm long and write where it ends. What do you notice? Start with: I notice ...

> **Model answer: The length stays 6 cm every time. From the 2 mark the line ends at 8; from 3 it ends at 9; from 5 it ends at 11. The end mark is always the start mark plus 6, so the end mark take away the start mark is always 6. Reading only the end mark gives the length only when the line starts at 0.**

Look for: A child who tries more than one start mark, writes both marks, and says that the end mark take away the start mark (or the number of jumps) is the length. A 15 cm ruler allows start marks up to 9.
Common mistakes: Writing the end mark as the length (8 for a line from 2 to 8).; Trying only one start mark.

Talk and explain (minutes 70 to 76)

Mei measures a pencil and says it is 15 cm. Ben measures the same pencil and says it is 16 cm. Why might their answers be different? How can we check? Sentence starter: I think ... because ...

> **Model answer: The tip may be between the 15 mark and the 16 mark, so one child chose the other mark. Or one child did not start at the 0 mark. We can check by measuring again with the 0 mark at the start, and by choosing the closer mark.**

Look for: A reason about the start at 0 or about an end between two marks, and a way to check (measure again). Praise the check, not who is right. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Draw a line 6 cm long on your spare paper. Start it at the 0 mark of your ruler. Then fill in the marks.**
It starts at the ___ mark and ends at the ___ mark.

Answer: A line from the 0 mark to the 6 mark

Worked solution: Put the 0 mark of the ruler on the dot and draw to the 6 mark. The line starts at 0 and ends at 6.
Common mistakes: Starting at the edge of the ruler.; Ending at the 7 mark after counting the 0 as one.

E2 **Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.**
The pencil is ___ cm longer than the rubber.

Answer: 9 cm longer

Worked solution: The rubber is 6 cm and the pencil is 15 cm. 15 take away 6 is 9, so the pencil is 9 cm longer.
Common mistakes: Adding: 21.; Writing 15 (the pencil, not the difference).

E3 **A line starts at the 3 mark and ends at the 10 mark. How long is the line?**
The line is ___ cm long.

Answer: 7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10)

Worked solution: Count the jumps from 3 to 10: 7 jumps. Or take 10 take away 3, which is 7. The line is 7 cm long, not 10 cm.
Common mistakes: Writing 10 (the end mark).; Writing 13 by adding.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (starting at the 0 mark; drawing a line exactly; a line that starts at another mark; adding the pieces of a path; something else); *What helped me* (my ruler; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my tutor will come and help)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Draw a line 8 cm long on spare paper. Start it at the 0 mark of your ruler. Which mark does it end at?	It starts at 0 and ends at 8
H2	Start here	A pencil starts at the 0 mark and its tip is at the 17 mark. A crayon is 11 cm long. How many cm longer is the pencil?	6 cm
H3	Start here	Measure a spoon, a toy and a crayon to the nearest cm. Write the 3 lengths in order, shortest first.	OPEN ANSWER, accept any three lengths, written from smallest to largest. A grown-up can check one by measuring again
H4	Think harder	A line starts at the 4 mark and ends at the 15 mark. How long is the line? Write: I know because ...	11 cm (15 take away 4)
H5	Go further	Draw a path with 3 straight pieces. The pieces are all different lengths. The whole path is 25 cm long. Write the 3 lengths. A grown-up checks with a ruler.	OPEN ANSWER, accept any 3 different whole numbers that add up to 25, for example 6, 8 and 11 (6 + 8 + 11 = 25)

Online practice (`out/l09/practice`, key `t1l09`): 13 items; 10 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to measure and draw lines to the nearest centimetre with a ruler and to say how they know. This week: starting at 0, reading the end mark, and comparing lengths.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–4	**Draw and Measure** s-title	TODAY WE WILL: / Use a ruler to measure and to draw lines exactly.	Lesson 9 of Term 1, 90 minutes. Objective: measure and draw lines to the nearest centimetre with a ruler. Time plan: opening 0 to 4, warm-up 4 to 10, sneak-peek 10 to 14, explore 14 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Tell the group the lesson is recorded (you and the screen only) and that a child may keep the camera off.
2	4–10	**Warm-up: measure with clips** s-rem1	Lay clips along your pencil, end to end. / No gaps. No overlaps. / Count the clips. / WHAT IF?: / Your friend uses bigger clips. Will the number be the same?	Children lay paper clips (or any same-size things) along a pencil and count them. This is a spiral review of measuring in equal steps (holiday topic) and it lets a new child join: they see how it is done before they try. Show on camera with big clips and then small clips: the same pencil gives two different numbers, so we need one unit that everybody shares. Say centimetre only at the end of the part.
3	4–10	**Find 0 on your ruler** s-rem2	Put your finger on the 0 mark. Count along the marks. / [picture: number_line] / Say the numbers with me	Every child puts a finger on the 0 mark of their own ruler and counts along with you: 0 to 10, then on to 12. The strip on screen shows the numbers only, the real ruler is the one the child holds. Quick questions: the mark after 4 is 5, the mark before 10 is 9, 12 comes after 10. Rulers with a gap before the 0 are normal: we start at the 0 mark, not at the edge.
4	4–10	**One centimetre** s-rem3	The gap between two marks is 1 cm. / [picture: number_line] / Slide your finger from 0 to 5. How many gaps? How many cm?	A centimetre (cm) is the gap from one mark to the next. From 0 to 5 is five gaps, so 5 cm. The 0 is where we start: it is not a centimetre. Answers to the quick questions: 0 to 8 is 8 cm, 0 to 10 is 10 cm. If a child says 6 for 0 to 5, they counted the marks: ask them to count the jumps.
5	10–14	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Draw a dot. / Use your ruler. / Draw a line from the dot, exactly 3 cm long. / PUZZLE 2: think: / Ali puts one end of a pencil at the 1 mark of his ruler. / The other end is at the 9 mark. / Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.	Puzzle 1 (a 3 cm line) is an access question: any child with a ruler can try it. Puzzle 2 is a thinking question on the lesson idea: where the pencil starts matters. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Everyone holds the paper to the camera once, at the same moment, after puzzle 2; there is no private chat and no board in front of the group. Answers: a line 3 cm long; no, 8 cm (the pencil starts at the 1 mark). Do not reveal Puzzle 2 until Talk time. Do not write scores or keep a list of who found it easy.
6	14–28	**Explore: guess, then measure** s-explore1	Find 3 things: a pencil, a spoon, a rubber or a toy. / Guess how many cm long each one is. / Put one end at the 0 mark. Read the mark at the other end. / Say: my ___ is about ___ cm. / WHAT IF?: / The end is between two marks. Which mark is closer?	Round 1, about 6 minutes. Objects first, little writing. Children guess, then measure three things from home. Ask a few children to hold the thing and the ruler to the camera: look at where the thing starts on the ruler (the 0 mark, not the edge). The what-if card is open to everyone from the start and no child is named for using it: it previews the nearest-centimetre idea in the mini-lesson. A child with a 15 cm ruler measures shorter things.

7	14–28	**Draw it!** s-explore2	Put a dot where the line starts. / Put the 0 mark on the dot. / Make a small mark at the end. / Join them with a straight line. / 3 cm 8 cm 12 cm	Round 2, about 5 minutes. Call each length and let children draw it. On a count of three they hold the paper to the camera: you can see where the lines start, not their exact length, so do not judge on camera. Then ask about Ravi, who started at the edge of the ruler and not at the 0 mark: his line is a little longer than 8 cm. Answers: any line drawn with the 0 mark at the start and the end mark at 3, 8 and 12.
8	14–28	**What did we notice?** s-explore3	Where do we start measuring? / What do we do when the end is between two marks? / Why do we all use centimetres?	Take one or two answers from volunteers. Key ideas: start at the 0 mark, not at the edge; read the mark at the other end; when the end is between two marks choose the closer mark, which is the nearest centimetre; everybody uses the same unit so everybody gets the same answer (the clips gave different numbers). Repeat a child's answer in your own words so the recording picks it up.
9	28–38	**The parts of a ruler** s-m1	[picture: number_line] / REMEMBER: / 0 is where we start. / The gap from one mark to the next is 1 cm.	Hold your real ruler to the camera next to this slide. The numbers sit on the marks; the centimetre is the gap between two marks. Ask: how many centimetres from the 0 mark to the 6 mark? Answer: 6. Say the word mark every time, and say gap or jump for the centimetre.
10	28–38	**Measure: three steps** s-m2	Put one end of the thing at the 0 mark. / Keep the ruler straight along the thing. / Read the mark at the other end. That is the length in cm. / Look from straight above	Show the three steps on camera with a pencil, one end exactly on the 0 mark. Say each step as you do it. Read the mark at the tip aloud, with the unit: 14 cm, for example. Looking from straight above matters: from the side the marks seem to shift. Do not ask a child to demonstrate yet.
11	28–38	**Draw a line 7 cm long** s-m3	Put a dot where the line starts. / Put the 0 mark on the dot. / Make a small mark at the 7. / Join them with a straight line. / Check: 0 at the start, 7 at the end.	Draw the line on paper under your camera while you say the five steps. Hold the ruler still with one hand and draw along the edge with the other. At the end, check aloud: the 0 mark was on the dot and the 7 mark is at the other end. A line that is a little long is the most common slip: it usually means the line started at the edge of the ruler.
12	28–38	**Between two marks** s-m4	The end is between the 8 mark and the 9 mark. It is closer to the 9 mark. / 8 9 / NEAREST CENTIMETRE: / Choose the closer mark. This one is about 9 cm.	Use a real pencil whose tip falls between two marks. When the tip is clearly closer to one mark, choose that mark: that is the nearest centimetre. Ask the second example: between 12 and 13, closer to 12: answer 12 cm. Today every example is clearly closer to one mark; do not set one exactly halfway.
13	28–38	**A line that starts at 3** s-m5	[picture: number_line] / The line starts at 3 and ends at 8. Count the jumps. 5 jumps. The line is 5 cm long. / 8 take away 3 is 5	A line from the 3 mark to the 8 mark: the numbers show six marks, but only five jumps between them. Count the jumps aloud: 1, 2, 3, 4, 5. The length is the end mark take away the start mark: 8 take away 3 is 5. It is not 8, because the first 3 cm of the ruler are not part of the line. Ask: why is it 5 cm and not 8 cm? Link back to puzzle 2 only after Talk time.
14	28–38	**Your turn** s-m6	Write on your paper. About 1 minute / DRAW: / Draw a line 5 cm long. / WORK OUT: / A line goes from the 2 mark to the 9 mark. / How long is it?	Each child draws and writes on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group. If a child finishes early, they may check the line by measuring it again, starting at a different mark.
15	28–38	**Check: your turn** s-m6a	DRAW: / The line starts at the 0 mark and ends at the 5 mark. / WORK OUT: / 9 take away 2 is 7. / 7 cm	Read each answer aloud so a child watching the recording can mark their own work. The 5 cm line: 0 at the start, 5 at the end. The second line is 7 cm: count 7 jumps from 2 to 9, or take 2 from 9. If a child wrote 9, ask them to count the jumps from the 2.
16	38–42	**Jump the tens, hop the ones** s-break	Pretend a big jump is 10 cm. A small hop is 1 cm. / 14 23 30 8 / 14 cm is 1 big jump and 4 small hops. Sip some water.	Four minutes. Stand up on camera. Call each number and have children jump the tens and hop the ones; count the jumps and hops aloud. No equipment. Keep it gentle for small spaces. The big jump for ten links to bundles of ten from Lesson 1, but do not teach that link now.

17	42–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. About 9 minutes. Use your ruler / questions S1, S2, S3	Read questions 1 and 2 aloud, then question 3. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. For question 3 a child may measure any pencil: the answer depends on the pencil. After question 3 any child may choose to carry on or try Think harder.
18	42–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group. Question 3 depends on the pencil: a guess within a few centimetres of the measure is fine, and a new pencil is usually between 17 and 19 cm.
19	42–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 7. About 9 minutes. Use your ruler / questions S4, S5, S6, S7	Read question 6 aloud: it is a spot-the-mistake. Read question 7 aloud as well: the tip is between two marks. Children may check any answer by measuring their own pencil against the ruler.
20	42–70	**Check page 2** s-ws2ans	answers S4, S5, S6, S7	Read each answer aloud so a child watching the recording can mark their own work. Question 6: no, her line ends at the 7 mark so it is 7 cm; she should draw on to the 8 mark. Never ask a child to show a wrong answer to the group.
21	42–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Our idea in a new setting / Go further (page 4) | Bigger puzzles / Investigation (page 5) | Does the start mark matter? / Check | Finish and check your own work / Everyone may choose. You can swap at any time.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
22	70–76	**Talk and explain** s-talk	MEI AND BEN MEASURE THE SAME PENCIL: / Mei says 15 cm. / Ben says 16 cm. / Why might their answers be different? How can we check? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: the tip may be between the 15 and the 16 mark and the two children chose different marks, or one child did not start at the 0 mark. We can check by measuring again with the 0 at the start and choosing the closer mark. Repeat each child's answer in your own words for the recording.
23	70–76	**Back to the sneak-peek** s-sneakans	[picture: number_line] / No! The pencil starts at the 1 mark, so it is 8 cm long: 9 take away 1.	Reveal puzzle 2 now. The pencil starts at the 1 mark and ends at the 9 mark: eight jumps, so 8 cm. Say: many of you wrote a good reason. Do not read out anyone's earlier answer.
24	76–82	**Exit ticket** s-exit	On your own. No looking back. About 3 minutes / questions E1, E2, E3	Children answer on page 5. After the time everyone holds the ticket to the camera at the same moment, so only the tutor sees it (no private chat). Stamp the inquiry journal.
25	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour. Question 1: the line starts at 0 and ends at 6. Question 2: 15 take away 6 is 9. Question 3: 10 take away 3 is 7.
26	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, check in during the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby); never a direct message. The Next time I will try row is optional; at the start of the next lesson you may say to the whole group: today you could try drawing it, using objects or checking your answer.

27	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Story Steps: Read It Twice / You need a pencil and paper.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (stamps, the look-back). Do not start anything new. If the whole class is ahead, show the optional menu slide and use the Spare-time menu from the tutor guide.
28	optional	**Go further: how many cuts?** s-opt	A strip of paper is 10 cm long. You may cut it at the 1 mark, the 2 mark and so on, but not at the ends. / How many places can you cut? What if the strip is 20 cm long?	Optional, only if you are on time. Answer: 9 places for the 10 cm strip (the marks 1 to 9), and 19 places for the 20 cm strip. We cannot cut at the 0 mark or the 10 mark, because that is the end of the strip. Take two answers and move on.
29	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | A line ends at the 9 mark. Is it always 9 cm long? / Make a puzzle | Draw a path with 3 straight pieces for your grown-up to measure / Teach it back | Show how to draw a line 9 cm long, in steps / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your ruler. Pause or rewind any time.*	Gather your ruler, workbook, pencil and paper. Press play when ready.
4:00	Remember warm-up	*Recording: join in! Lay clips along a pencil, find 0 on your ruler, and count the marks out loud. Pause if you need longer.*	Play along. Do each step with your own ruler before the video does.
10:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your own paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
14:00	Explore	*Recording: pause now. Guess, then measure three things. Then draw lines of 3, 8 and 12 centimetres. Press play when you have tried.*	Pause. Guess and measure three things, then draw each line with your ruler. Write what you find.
28:00	Mini-lesson	*Recording: keep watching and copy each example with your own ruler. Rewind if you miss a step.*	Watch and copy the examples with your ruler. Rewind any step you missed.
38:00	Brain break	*Recording: stand up and join in! Big jump for ten centimetres, small hop for one.*	Stand up and jump the tens and hop the ones.
42:00	Workbook	*Recording: pause now and do workbook page 1 with your ruler. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*	Pause. Explain your answer to someone at home, with a reason.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own, with your ruler. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.

85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	4:00	2:58	3:27	ok
remember	6:00	5:06	6:00	ok
sneak	4:00	3:42	4:20	ok
explore	14:00	11:42	13:55	ok
mini	10:00	8:31	10:02	ok
break	4:00	3:32	4:12	ok
work	28:00	26:42	31:50	ok
talk	6:00	5:27	6:28	ok
exit	6:00	4:28	5:17	ok
look	3:00	2:28	2:55	ok
flex	5:00	0:38	0:44	ok
Total	90:00	75:13	89:08	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 7 questions; a middle child needs about 8.5 min for page 1 and 9.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.